

Report
Of
Java Development Internship at Codec Technologies Pvt. Ltd.

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
MCA



By

Roll Number of Student: 25SCS2040005589

Name of Student: ANKIT YADAV

Batch: 2025-27

CANDIDATE'S DECLARATION

I, Ankit Yadav, Roll No. 25SCS2040005589, do hereby declare that the internship report titled “Java Development Internship at Codec Technologies Pvt. Ltd.” has been completed by me to fulfil the requirement for the award of the degree of Master of Computer Applications (MCA) at IILM University, Greater Noida. All the references used in this report have been properly quoted, and I have not copied any material from any other source without acknowledgement. I confirm that this report is my own original work and has not been submitted earlier to any other institute for any degree or diploma. I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: Ankit Yadav

Roll No.: 25SCS2040005589

ACKNOWLEDGEMENT

I would like to use this opportunity to thank everyone who supported and guided me during my internship at Codec Technologies Pvt. Ltd. I am truly thankful for their kind guidance, helpful suggestions, and constant encouragement throughout this internship.

I sincerely thank Dr. Anurag Shrivastava and the Codec Technologies team for giving me the opportunity to work as a Java Developer Intern, and for their continuous support, mentorship, and feedback throughout the internship period. I am also thankful to the National Internship Portal (AICTE) for providing a platform through which this AICTE & ICAC approved internship was made available to students.

I am also grateful to my internal project guide and the faculty members of the School of Computer Science and Engineering, IILM University, Greater Noida, for their valuable guidance and encouragement during this internship.

Finally, I express my heartfelt gratitude to my parents and family members for their constant support, blessings, and motivation throughout this journey.

Signature

Student Name: Ankit Yadav

Roll No.: 25SCS2040005589

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INTERNSHIP OFFER LETTER

This chapter contains the official Internship Offer Letter issued by Codec Technologies Pvt. Ltd., through the National Internship Portal, confirming the internship placement, role, and duration.



INTERNSHIP OFFER LETTER

Dear ANKIT YADAV,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Java Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 15/09/2026 to 15/10/2026
- Reporting to : Assigned Project Head(s)

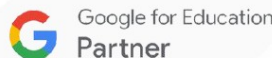
This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,
Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 15/09/2026


Dr. Anurag Shrivastava
Talent Acquisition Manager



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Internship Offer Letter (Date of Issue: 15/09/2026)

CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents the work I carried out as a Java Developer Intern at Codec Technologies Pvt. Ltd. The internship was a 1 Month AICTE & ICAC approved program provided through the National Internship Portal, with a duration from 15 September 2026 to 15 October 2026. It was conducted in a hybrid mode under the guidance of the assigned Project Head(s).

During the internship, I built an Employee Management System using Java, the Spring MVC framework, Hibernate, and MySQL. Through this project, I learned how a complete enterprise-style application is structured, how data is stored and retrieved from a database, and how different users can be given different levels of access. This report explains the company, the problem, the tools, the steps followed, and what I learned.

4.2 Organization Profile

Codec Technologies is an IT and business consultancy company with its office at Chandivali IT Hub, Mumbai. As stated in the offer letter, the company delivers IT and business consultancy services in more than 27 countries and acts as a global platform that connects learners with talent and career opportunities. It is ISO 9001:2015 certified and is a Google for Education Partner.

Through the National Internship Portal, the company offers internship programs that give students industry-level knowledge. Interns are given project tasks that are completed on the company's platform and through email, and their progress is reviewed every week by the project supervisor. Exceptional interns may also qualify for a scholarship award.

4.3 Problem Statement

Many organizations still keep employee details in scattered files such as spreadsheets and paper records. This makes it hard to find information, easy to create duplicate or wrong entries, and difficult to control who is allowed to see or change sensitive details like salary or personal data. As the number of employees grows, these problems become bigger.

To solve this, I worked on an Employee Management System, a centralized application where employee records can be added, updated, deleted, and viewed in an organized way. The system also uses role-based access, so that each user can only perform the actions allowed for their role.

4.4 Project Objectives

- To learn how to build a web application in Java using the Spring MVC framework.
- To implement complete CRUD operations (Create, Read, Update, Delete) for employee records.
- To implement role-based access so that different users get different permissions.

- To understand object-relational mapping by using Hibernate to connect Java classes with MySQL tables.
- To design a clean MySQL database for storing employee information.
- To follow professional working habits such as weekly reviews, regular updates, and clear communication.

4.5 Scope of the Project

The internship focused on one main project:

- **Employee Management System:** An application to manage employee records using CRUD operations. Users can add, update, delete, and view employee data, and the actions available to each user depend on their role.

The scope was limited to designing and developing this application as assigned during the internship period. It did not include changes to any live company system or the use of any real employee data.

4.6 Technologies and Tools Used

- **Java** – the core programming language used to write the whole application.
- **Spring MVC** – the framework used to organize the application into controllers, services, and views, and to handle web requests.
- **Hibernate** – an ORM (Object-Relational Mapping) tool that lets the Java code save and read data from the database without writing long SQL by hand.
- **MySQL** – the relational database used to store employee records and user roles.

4.7 System Architecture

The application follows the MVC (Model-View-Controller) pattern, which separates the application into clear layers. The flow of a request is as follows:

- **Step 1 – View:** The user opens a web page and performs an action, such as filling a form to add a new employee.
- **Step 2 – Controller:** Spring MVC receives the request and passes it to the correct controller.
- **Step 3 – Service Layer:** The business rules are applied here, including checking whether the user's role allows the action.
- **Step 4 – Data Access Layer:** Hibernate converts the Java objects into database operations and communicates with MySQL.
- **Step 5 – Response:** The result is sent back through the controller and shown on the page, for example an updated employee list.

4.8 Methodology

Project: Employee Management System

Description: Create an application to manage employee records (CRUD operations).

What I Did: I built the application in Java using Spring MVC, with Hibernate for database access and MySQL for storage. I designed the employee table and mapped it to a Java entity class, then created the controllers and services needed for the four main operations: adding a new employee, updating existing details, deleting a record, and viewing employee data. I also added role-based access so that users with different roles can perform only the actions permitted to them.

Goal: To understand how a real business application is built with a layered structure and how data and access rules are managed in it.

Skills Learned: Java programming, Spring MVC (controllers, services, and views), Hibernate mapping and database operations, MySQL table design, and implementing role-based access control.

4.9 Weekly Work Plan

The internship lasted four weeks. The work was divided into weekly phases so that progress could be checked in the weekly performance reviews by the project supervisor.

Week	Phase	Main Activities
Week 1	Orientation and Setup	Understanding the internship program and project requirements; setting up Java, the IDE, and MySQL; learning the basics of Spring MVC and Hibernate; planning the database.
Week 2	Core Development	Creating the database tables and Java entity classes; building the controllers and services; implementing the add and view employee features.
Week 3	Feature Completion	Implementing update and delete operations; adding role-based access; connecting all layers and improving the pages.
Week 4	Testing and Submission	Testing every operation and role, fixing bugs, cleaning the code, preparing documentation, and submitting the final work and report.

4.10 Detailed Modules and Features

The table below explains the main modules of the Employee Management System.

Module	Purpose
Add Employee	Allows an authorized user to enter the details of a new employee and save them in the database.
Update Employee	Allows changes to an existing employee's details, so records stay correct and up to date.
Delete Employee	Removes an employee record that is no longer needed.
View Employee Data	Displays the list of employees and the details of each one.
Role-Based Access	Controls which actions each type of user can perform, protecting sensitive data from unauthorized changes.
Database Layer	Uses Hibernate and MySQL to store, retrieve, and manage employee information safely.

4.11 Challenges Faced and Solutions

Every project has some difficulties. The table below explains the main challenges I faced and how I handled them.

Challenge	How I Handled It
Understanding the structure of Spring MVC	Studied the official documentation and built small sample controllers before starting the main project.
Mapping Java classes to database tables with Hibernate	Started with one simple entity, tested saving and reading data, and then extended it to the full employee record.
Restricting actions according to user roles	Planned the roles first and checked the role in the service layer before allowing any add, update, or delete action.
Fixing errors between the layers	Tested each layer separately, which made it easier to find where an error was happening.
Completing the work within one month	Followed the weekly plan, focused on core CRUD features first, and asked the project head when I was stuck.

4.12 Learning Outcomes

This internship helped me grow in both technical and professional areas:

- **Java development:** writing clean, structured Java code for a complete application.
- **Framework knowledge:** understanding how Spring MVC organizes an application into controllers, services, and views.
- **Database skills:** using Hibernate and MySQL together and understanding how objects are stored in tables.
- **Security awareness:** learning why role-based access is important for protecting employee information.

- **Professional habits:** meeting weekly targets, taking feedback from the project head, and delivering work on time.

4.13 Expected Outcomes

- A working Employee Management System that can add, update, delete, and view employee records.
- A role-based access feature that limits actions according to the user's role.
- Practical experience with Java, Spring MVC, Hibernate, and MySQL.
- Better understanding of layered application design and database handling.
- More confidence in building and testing a complete application on my own.

4.14 Future Scope

The Employee Management System can be extended in many ways. Future improvements can include attendance and leave management, payroll and salary slip generation, search and filter options, reports and dashboards for HR teams, and secure login with password encryption. The system can also be exposed through REST APIs so that it can be used by a mobile application.

4.15 Conclusion

The one-month Java Developer Internship at Codec Technologies was a valuable and practical learning experience. By building an Employee Management System with Java, Spring MVC, Hibernate, and MySQL, I learned how a real application is planned, developed, tested, and improved. I also gained confidence in working with a layered architecture and in handling data and user roles safely.

I thank Codec Technologies, the National Internship Portal, and my university for giving me this opportunity. The skills I have gained will help me in my studies and in my future career as a software developer.

4.16 Certificates of Completion and Other Communication Proof

The Internship Offer Letter (dated 15/09/2026, NCS ID: E19E86-0116588288923) issued by Codec Technologies Pvt. Ltd. is attached in Chapter 3 of this report as proof of the internship. It was issued through the National Internship Portal. The Internship Completion Certificate (AICTE ID: CORPORATE6759d549ce59e1733940553), once received, may be attached here separately.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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